$$\frac{\partial t}{\partial \tau}=\frac{\lambda}{\rho c}\left(\frac{\partial^2 t}{\partial x^2}+\frac{\partial^2 t}{\partial y^2}+\frac{\partial^2 t}{\partial z^2}\right)+\frac{\dot{\Phi}}{\rho c}$$

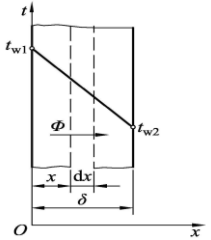
- 第一类
- 第二类
- 第三类

$$\Phi=-A\lambda\frac{\mathrm{d}t}{\mathrm{d}x}$$

$\lambda_{固相}>\lambda_{液相}>\lambda_{气相}$

$\lambda_{金属}>\lambda_{非金属}$

稳态



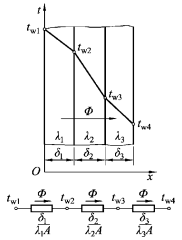
$$\cancel{\frac{\partial t}{\partial \tau}}=\frac{\lambda}{\rho c}\left(\frac{\partial^2 t}{\partial x^2}+\cancel{\frac{\partial^2 t}{\partial y^2}}+\cancel{\frac{\partial^2 t}{\partial z^2}}\right)+\cancel{\frac{\dot{\Phi}}{\rho c}}$$

- 第一类

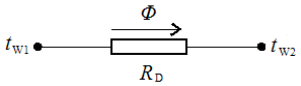
$$q=\frac{\lambda(t_{w1}-t_{w2})}{\delta}=\frac{\lambda\Delta t}{\delta}\qquad \mathrm{W/m^2}$$

$$\Phi=Aq=\frac{\lambda A(t_{w1}-t_{w2})}{\delta}=\frac{\lambda A\Delta t}{\delta}\qquad \mathrm{W}$$

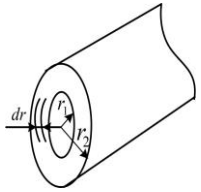
$$t(x)=t_{w1}-\frac{t_{w1}-t_{w2}}{\delta}x$$



$$q=\frac{t_{w1}-t_{w4}}{\frac{\delta_1}{\lambda_1}+\frac{\delta_2}{\lambda_2}+\frac{\delta_3}{\lambda_3}}$$

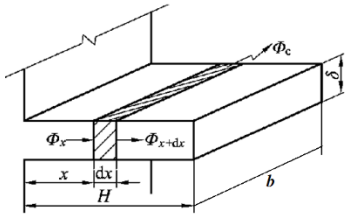


单位管长热流量



$$q_l=\frac{\Phi}{l}=\frac{t_{w1}-t_{w2}}{\frac{1}{2\pi\lambda}\ln\frac{d_2}{d_1}}\qquad \mathrm{W/m}$$

$$R_{\lambda l}=\frac{1}{2\pi\lambda}\ln\frac{d_2}{d_1}\qquad (\mathrm{m\cdot ℃/W})$$



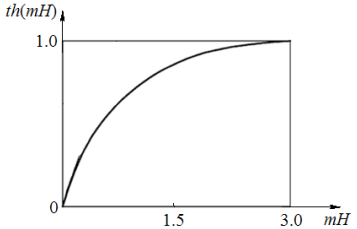
$$\cancel{\frac{\partial t}{\partial \tau}}=\frac{\lambda}{\rho c}\left(\frac{\partial^2 t}{\partial x^2}+\cancel{\frac{\partial^2 t}{\partial y^2}}+\cancel{\frac{\partial^2 t}{\partial z^2}}\right)+\frac{\dot{\Phi}}{\rho c}$$

- 第三类

$$\theta=\theta_0\frac{ch[m(H-x)]}{ch(mH)}\qquad \theta_H=\theta_0\frac{1}{ch(mH)}\qquad m^2=\frac{hU}{\lambda A}$$

$$\eta_f=\frac{\Phi}{\Phi_0}\qquad \eta_f=\frac{\Phi}{\Phi_0}=\frac{th(mH)}{mH}$$

$$\Phi=\eta_f\Phi_0=\eta_f hUH\theta_0$$

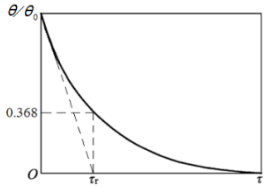


非稳态

$$\frac{\partial t}{\partial \tau}=\frac{\lambda}{\rho c}\left(\cancel{\frac{\partial^2 t}{\partial x^2}}+\cancel{\frac{\partial^2 t}{\partial y^2}}+\cancel{\frac{\partial^2 t}{\partial z^2}}\right)+\frac{\dot{\Phi}}{\rho c}$$

- 第三类

$$\theta=\theta_0 e^{-\frac{hA}{\rho cV}\tau}$$



$$Bi_V=\frac{hL}{\lambda}\qquad \mathrm{Bi}\le 0.1\qquad L=R$$

$$\tau_r=\frac{\rho cV}{hA}\qquad \frac{\theta}{\theta_0}=\exp(-\mathrm{Bi}_V\cdot \mathrm{Fo}_V)\qquad Fo_V=\frac{a\tau}{L^2}\qquad \mathrm{Bi}_V\le 0.1M\qquad L=V/A$$

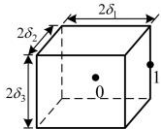
$$\frac{\partial t}{\partial \tau}=\frac{\lambda}{\rho c}\left(\frac{\partial^2 t}{\partial x^2}+\frac{\partial^2 t}{\partial y^2}+\frac{\partial^2 t}{\partial z^2}\right)+\frac{\dot{\Phi}}{\rho c}$$

- 第三类

$$\cancel{\frac{\partial t}{\partial \tau}}=\frac{\lambda}{\rho c}\left(\frac{\partial^2 t}{\partial x^2}+\cancel{\frac{\partial^2 t}{\partial y^2}}+\cancel{\frac{\partial^2 t}{\partial z^2}}\right)+\cancel{\frac{\dot{\Phi}}{\rho c}}$$

$$\frac{\theta(x,\ \tau)}{\theta_0}=f\left(Fo、Bi、\frac{x}{\delta}\right)$$

$$\frac{\theta(x,y,z,\tau)}{\theta_0}=\frac{\theta(x,\tau)}{\theta_0}\cdot\frac{\theta(y,\tau)}{\theta_0}\cdot\frac{\theta(z,\tau)}{\theta_0}$$

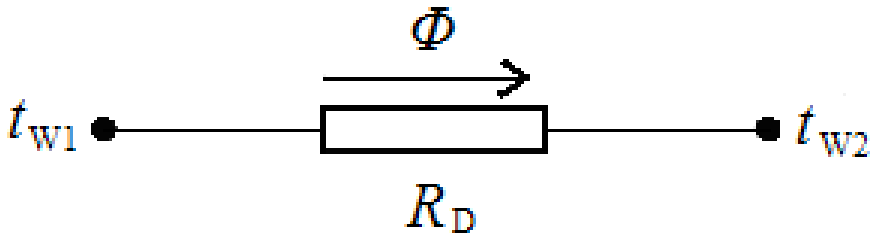


$$\begin{cases}\dfrac{\theta_m}{\theta_0}=\varphi(Fo、Bi)\\[2ex] \dfrac{\theta}{\theta_m}=\psi\left(Bi、\dfrac{x}{\delta}\right)\end{cases}$$

用热电偶监测气流温度随时间变化规律时，应如何选择热电偶节点的大小？

重点1：平板或圆筒导热过程的计算

$$q=\frac{\lambda(t_{w1}-t_{w2})}{\delta}=\frac{\lambda\Delta t}{\delta} \qquad \mathrm{W/m^2}$$



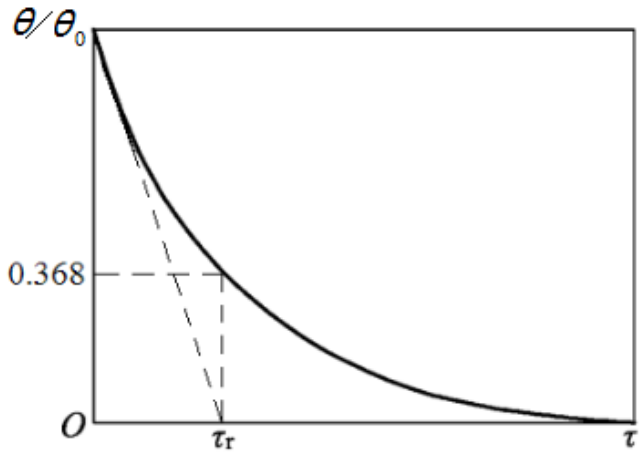
$$q_l=\frac{t_{w1}-t_{w,n+1}}{\sum_{i=1}^{n}\frac{1}{2\pi\lambda_i}\ln\frac{r_{i+1}}{r_i}} \qquad \mathrm{W/m}$$

例题

由三层材料组成的加热炉炉墙。第一层为耐火砖。第二层为硅藻土绝热层，第三层为红砖，各层的厚度及导热系数分别为$\delta_1=240mm$，$\lambda_1=1.04W/(m\cdot℃)$，$\delta_2=50mm$，$\lambda_2=0.15W/(m\cdot℃)$，$\delta_3=115mm$，$\lambda_3=0.63W/(m\cdot℃)$。炉墙内侧耐火砖的表面温度为1000℃。炉墙外侧红砖的表面温度为60℃。试计算硅藻土层的平均温度及通过炉墙的导热热流密度。

重点2：集总参数法

$$\frac{\theta}{\theta_0} = \exp\left(-\frac{hA}{\rho c V}\tau\right)$$



3、25^{0}C的热电偶被置于温度为250^{0}C的气流中，设热电偶节点可以近似看成球形，要使其时间常数$\tau_c = 1s$，问热节点的直径为多大？忽略热电偶引线的影响，且热节点与气流间的表面传热系数为**h=300W /(m^2 K)**，热节点材料的物性参数为：$\lambda = 20W/(mK)$，$\rho = 8500kg/m^3$，$c = 400J/(kg \cdot K)$ 如果气流与热节点间存在着辐射换热，且保持热电偶时间常数不变，则对所需热节点直径大小有何影响？

$$\pi d_i l h(t_w - t_f) = \frac{\pi}{4} d_i^2 u_f \rho_f c_p (t_f'' - t_f')$$

$$\Phi = hA(t_f - t_w) \qquad \mathrm{W}$$

$$q = h(t_f - t_w) \qquad \mathrm{W/m^2}$$

$$\Delta t_m = \frac{1}{2}(\Delta t' + \Delta t'') = t_{w,m} - t_{f,m}$$

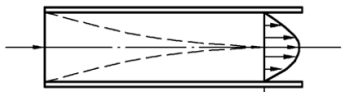
$$\Delta t_m = \frac{\Delta t' - \Delta t''}{\ln \frac{\Delta t'}{\Delta t''}} = \frac{\Delta t_{max} - \Delta t_{min}}{\ln \frac{\Delta t_{max}}{\Delta t_{min}}}$$

$$Nu = \frac{hL}{\lambda}$$

$$Nu_f = 0.023 Re_f^{0.8} Pr_f^{0.4} c_t c_l c_R$$

$$Pr = \frac{\mu c_p}{\lambda} = \frac{\nu}{a}$$

$$Re = \frac{\rho u L}{\mu}$$



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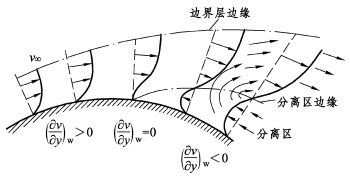
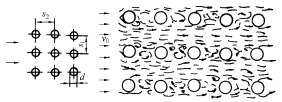




四种边界层

速度

$$u_f = \frac{\rho_1}{\rho_f} u_1 = \frac{T_f}{T_1} u_1$$

$$u_f = \frac{\rho_1}{\rho_f} u_1 = \frac{T_f}{T_1} u_1$$

主流速度

管间最大流速 $v'_{max} = \frac{v_0 s_1}{s_1 - d_o}$ $\quad v_{max} = \frac{v_0 s_1}{2(s_2' - d_o)}$

温度

$$t_f = \frac{1}{2}(t_f' + t_f'')$$

$$t_m = \frac{1}{2}(t_w + t_\infty)$$

$$t_m = \frac{1}{2}(t_w + t_\infty)$$

入口温度

雷诺数

$Re < 2300$ $\quad Re > 10^4$

$Re_l < 5 \times 10^5$

$Re < 1.4 \times 10^5$

[redacted]

长度

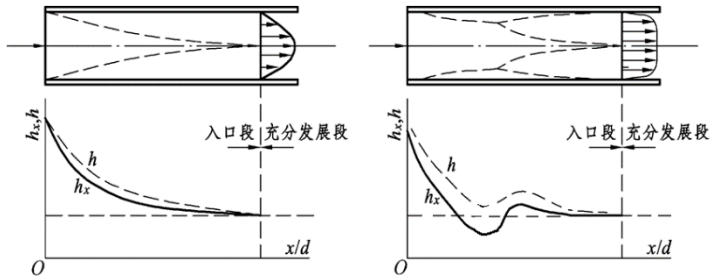
内径

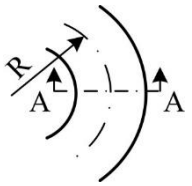
板长 l

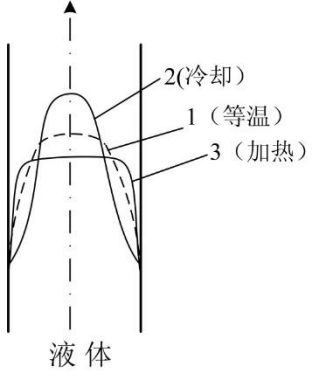
圆管外径 d_o

圆管外径 d_o

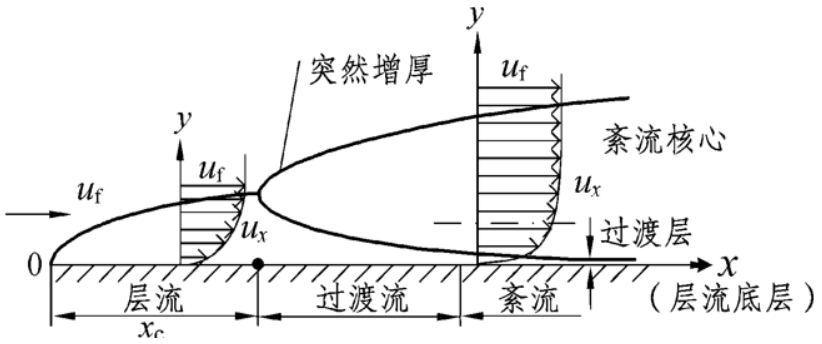
四种边界层





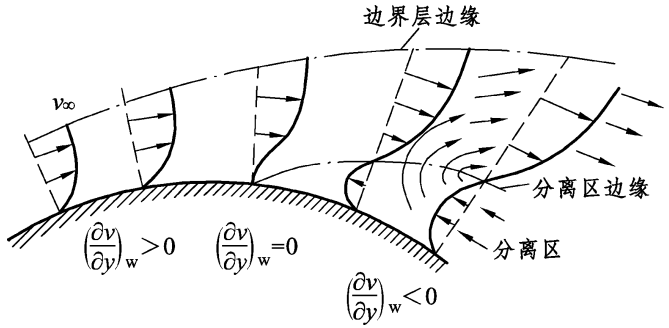


入口段效应
弯管
温度

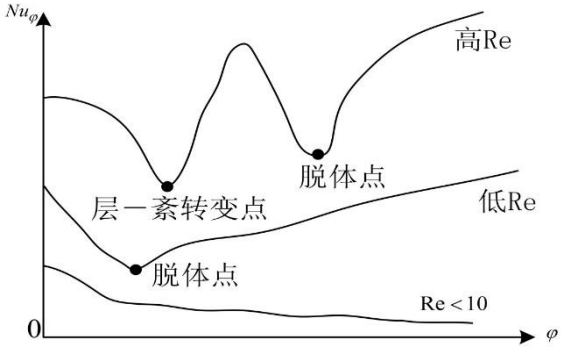


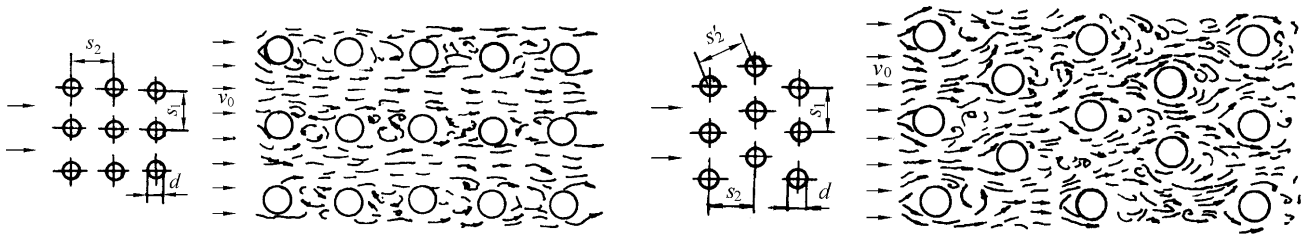
$$Re_{cm} = \frac{u_\infty x_c}{\nu_m}$$

$$Re_{cm} = 5\times10^5$$



- Re <1.4×10⁵ 时：$\varphi = 80°\sim85°$
- Re ≥1.4×10⁵ 时：$\varphi\approx140°$
- 低 Re（Re < 10）时：不会出现分离现象。



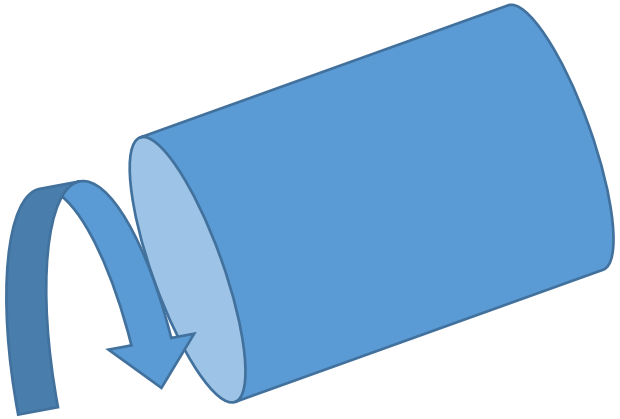


- ***Re*** 较低时，叉排前几排管子的 h_z 大于顺排，经过整个管子的平均 ***h*** 也大于顺排；
- ***Re*** 较高时，顺排管束的尾部涡流增强，使后排管子受到尾流影响的面积增加，而且由于涡流增强，扰动更强烈，以致顺排管束的 ***h*** 与叉排相当甚至可超过叉排管束。

式表明，在边界上垂直于壁面的热量传递完全依靠导热，那么在对流换热过程中流体的流动起什么作用

$$h = -\frac{\lambda}{\Delta t}\frac{\partial t}{\partial y}\bigg|_{y=0}$$

夏天，常常将饮料容器置于冰水中来冷却饮料。为了加速冷却，有人提出了这样一个专利：将饮料壳体（例如易拉罐）绕其轴线在冰水中做转动。如果能实现饮料瓶或易拉罐绕其轴线的纯转动，试从对流传热基本方程出发，分析这样的方法能否加速饮料的冷却？



重点3：对流换热过程的计算

- 特别注意特征温度，速度的定义

- 雷诺数的分界

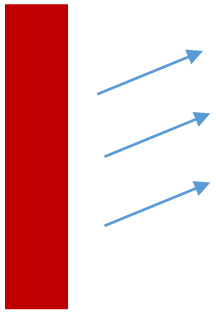
$$Nu_{\mathrm{f}} = 0.023 Re_{\mathrm{f}}^{0.8} Pr_{\mathrm{f}}^{0.4} c_t c_l c_R$$

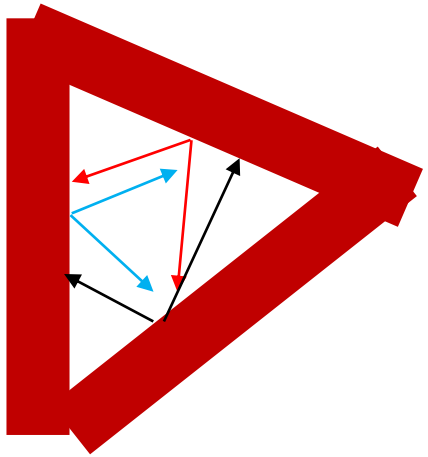
$$\pi d_i l h (t_{\mathrm{w}} - t_{\mathrm{f}}) = \frac{\pi}{4} d_i^{\,2} u_{\mathrm{f}} \rho_{\mathrm{f}} c_p (t_{\mathrm{f}}'' - t_{\mathrm{f}}')$$

6－14、已知：$1.013\times10^{5}\ Pa$ 下的空气在内径为 76mm 的直管内流动，入口温度为 65℃，入口体积流量为 $0.022m^{3}/s$，管壁的平均温度为 180℃。

求：管子多长才能使空气加热到 115℃。

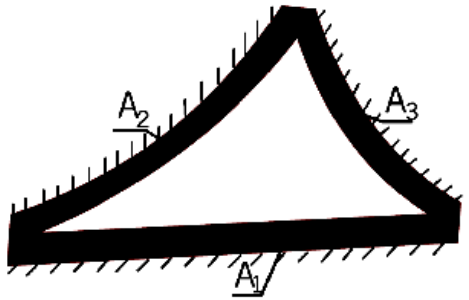
- 单物体辐射
  - 黑体辐射
    - 四次方定律
    - 普朗克定律
    - 兰贝特定律
  - 基尔霍夫定律
  - 灰体辐射
  - 角系数
- 热阻分析
- 基尔霍夫
- 多物体辐射





在不同光源的照耀下，物体常呈现不同的颜色，如何解释？

重点4：角系数的计算



$$A_1 X_{1,2} = A_2 X_{2,1}$$

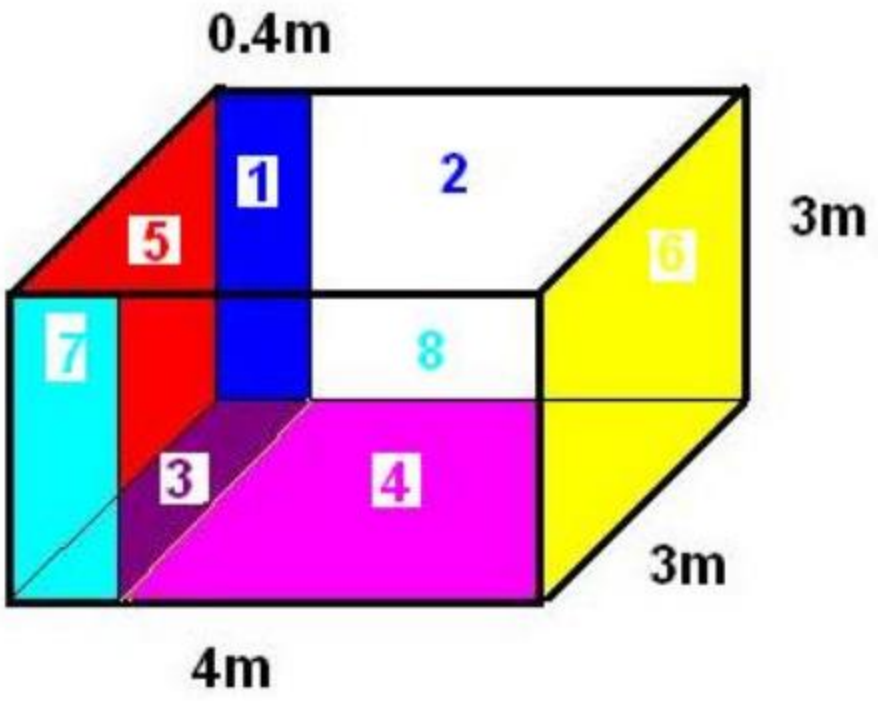
$$\sum_{j=1}^{n} X_{i,j} = X_{i,1} + X_{i,2} + \cdots + X_{i,n} = 1$$

$$X_{1,2} = \frac{A_1 + A_2 - A_3}{2A_1}$$

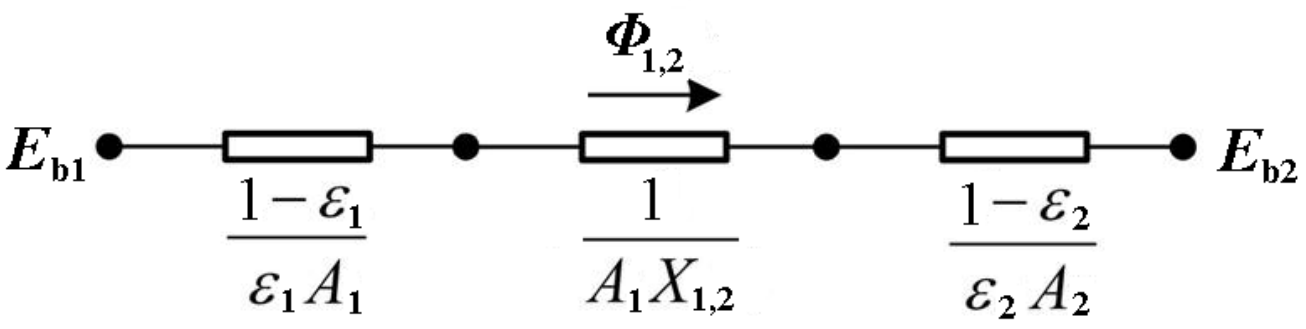
$$A_1 X_{1,2} = A_3 X_{3,2} + A_4 X_{4,2}$$

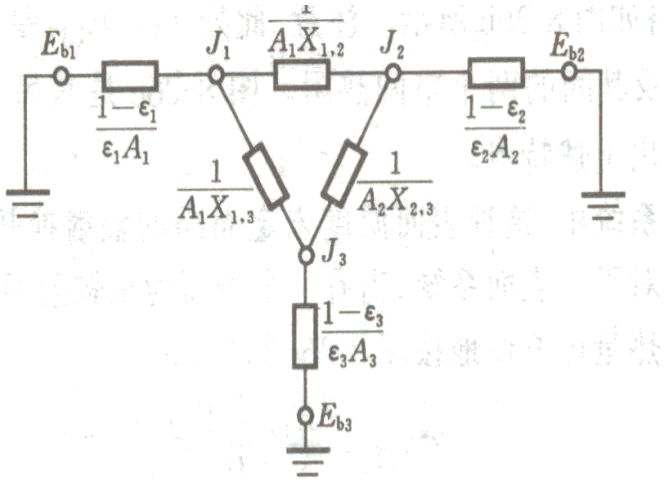
$$X_{1,2} = X_{1,5} + X_{1,6}$$

P252 .11 某采暖房间采用立式悬挂辐射采暖板，　试求此采暖板和房间各表面间的角系数，房间和采暖板的尺寸见图9－38。

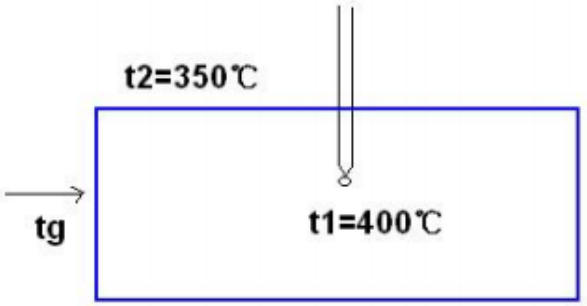


重点5：辐射传热过程的计算





p254.26用热电偶来测量管内流动着的空气的温度，如图9－39。热电偶读得温度t1＝400℃，管壁测得温度t2＝350℃，热电偶头部和管壁的发射率分别为0.8和0.7。从气流到热电偶头部的对流表面传热系数为35W/(m²K).试计算由于热电偶头部和管壁之间辐射换热而引起的测温误差，此时气流的真实温度是多少？讨论测温误差和换热系数的关系，测温误差和热电偶头部发射率的关系。



本章重点：

传热系数

$$\Phi = kA\Delta t = kA(t_{f1} - t_{f2})$$

$$k = \frac{1}{r_k} = \frac{1}{\frac{1}{h_1} + \sum_{i=1}^{n}\frac{\delta_i}{\lambda_i} + \frac{1}{h_2}}$$

$$k = \frac{1}{\frac{d_2}{d_1 h_1} + \sum \frac{d_2}{2\lambda_i}\ln\frac{d_{i+1}}{d_i} + \frac{1}{h_2}}$$

$$k_1 = \frac{1}{\frac{1}{h_1} + \frac{\delta}{\lambda} + \frac{1}{h_2 \eta_t \beta}}$$

$\beta = \frac{A_2}{A_1}$ —— 肋化系数

$\eta_t = (A_2'' + \eta A_2')/A_2$ —— 肋面总效率

临界热绝缘直径

$$d_c = \frac{2\lambda_{ins}}{h_2}$$

换热器的热计算

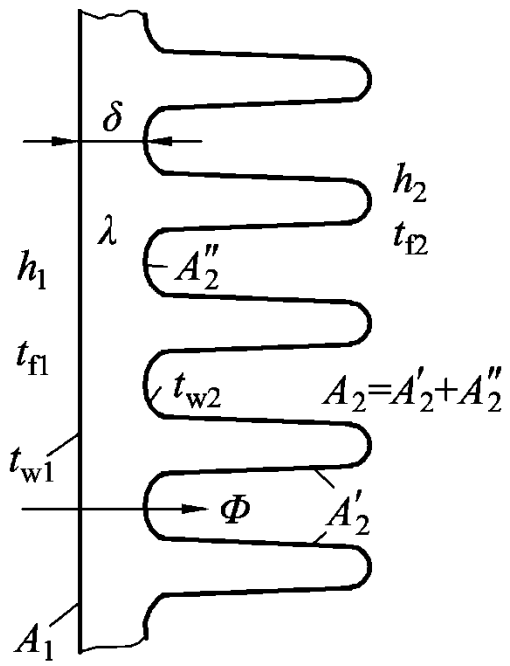
$$\Phi = kA\Delta t_m$$

$$\Phi = q_{m1}c_{p1}(t_1' - t_1'') = q_{m2}c_{p2}(t_2'' - t_2')$$

$$\Delta t_m = \frac{\Delta t_{max} - \Delta t_{min}}{\ln\dfrac{\Delta t_{max}}{\Delta t_{min}}}$$

重点6：传热过程计算

$$k_1=\frac{1}{\frac{1}{h_1}+\frac{\delta}{\lambda}+\frac{1}{h_2\eta_t\beta}}$$



$$\Phi=k_1A_1(t_{f1}-t_{f2})=k_2A_2(t_{f1}-t_{f2})$$

效率— 肋面总效率-肋化系数

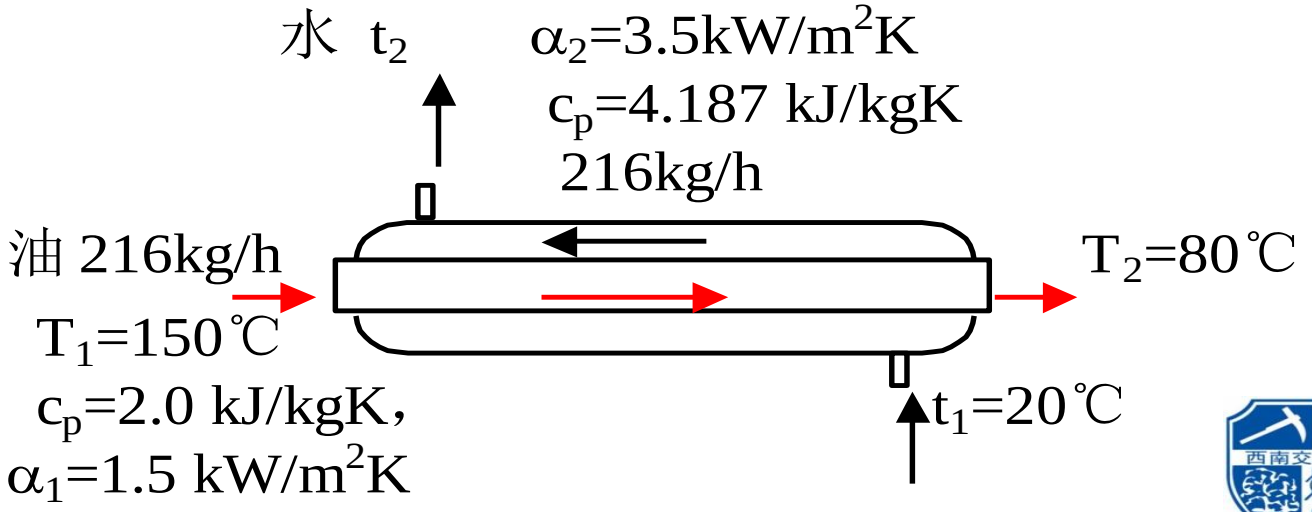
重点7：换热器计算

$$\Phi = kA\Delta t_{\mathrm{m}}$$

$$\Phi = q_{\mathrm{m}1}c_{p1}(t_1' - t_1'') = q_{\mathrm{m}2}c_{p2}(t_2'' - t_2')$$

$$\Delta t_{\mathrm{m}} = \frac{\Delta t_{\max} - \Delta t_{\min}}{\ln \dfrac{\Delta t_{\max}}{\Delta t_{\min}}}$$

例1 在套管式油冷却器里，热油在ϕ25×2.5mm 的金属管内流动，冷却水在套管环隙内流动，油和水的质量流量皆为 216kg/h，油的进、出口温度分别为 150℃和 80℃，水的进口温度为 20℃。油侧对流传热系数为 $1.5\,kW/m^2\cdot K$，水侧的对流传热系数为 $3.5\,kW/m^2\cdot K$，油的比热为 $2.0\,kJ/kg\cdot K$，试分别计算逆流和并流操作所需要的管长。忽略污垢热阻及管壁导热热阻。



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